

Abhimanyu Bhola

Date of Birth: 21 December 2000 | **Place of Birth:** Chandigarh, India | **Nationality:** Indian
Email: abhimanyubhola@gmail.com | **Phone:** +91- 978 169 2601 | **Portfolio:** abhimanyubhola.github.io

EDUCATION

- **École Polytechnique Fédérale de Lausanne (EPFL)**, Lausanne, Switzerland *Sep 2026- Present*
Master of Science, Microengineering
- **Indian Institute of Technology Madras**, Chennai, India *July 2018- July 2023*
Dual Degree Program – B. Tech in Engineering Design & M. Tech in Automotive Engineering
 - CGPA: 9.01/10
 - Graduated with the **2nd rank** in the class
 - Relevant Courses: Mechatronics System Design, Design of Mechanical Systems, Physics of Measurement, Control Systems, Smart Sensing for IoT, Pattern Recognition and Machine Learning, Analog and Digital Electronics

WORK EXPERIENCE

- **Break** *Nov 2025- Aug 2026*
 - Took a break to focus on my health and volunteer as a Math teacher for Grade 8 and 9 students in an NGO in Mumbai
- **Design Engineer** – Inito Health Tech Pvt Ltd, Bengaluru, India |  *July 2023- Nov 2025*
Patent (Filed): System and Method for Imaging a Test Strip, Indian Patent Office Application No. 202511033059
 - **Inventors:** Varun Akur Venkatesan, Purushottam Kadoo, **Abhimanyu Bhola**, Gautam Sharma, Siraj Issani, Rajesh Katreddy, Omkar Sanjay Karve
 - An **imaging system** used to capture and analyse images from test strips, and provide quantitative hormonal insights**Product Design**
 - Led end-to-end design and validation of **7 components** in a test strip imaging system, in collaboration with data science and firmware teams, achieving **95% accuracy** in hormone measurements against laboratory benchmarks
 - Designed and conducted **20+ experiments** during the development cycle to test and validate multiple designs based on **Blender Simulations**, using **SLA 3D printing** for prototyping and scaling up to **injection moulding** for production
 - Improved reliability and precision of the device, **achieving detection of 2× fainter signals** compared to the older generation; while also incorporating a **UV LED to support Europium-based assays**
 - Performed a detailed Design **Failure Modes and Effects Analysis (DFMEA)** on the imaging system, singling out key mechanical failure modes in the system; Made design changes for the highest risk failure modes**Production, Quality and Vendor Management**
 - Scaled up the in-house production **from 0 to a stable 300 units per week** by developing **assembly and quality control (QC) protocols**, and trained the quality and production engineers in these protocols
 - Developed an **imaging test target for the End-of-Line QC** to ensure each device's reliability by experimenting with technologies such as **photolithography, and screen, pad, and digital printing technologies**
 - Conducted on-site evaluations at 4 vendors across China to validate production of critical optical components including **cameras, first-surface mirrors, and beamsplitters**
- **Battery Engineering Intern** – Ather Energy, Bengaluru, India |  *Dec 2021 – May 2022*
 - Researched and conducted trials of laser welding for cell to busbar linkages by analysing **7 materials and 12 parameters** to meet pull force and resistance specifications
 - Developed Python scripts to characterize and measure battery pack imbalances of Ather battery packs in the field, and performed an RCA of >15 imbalance battery packs by working with the cell engineering team

RESEARCH EXPERIENCE

- **Undergraduate Researcher** – Virtual Proving Ground and Simulations Lab, IITM |  *Jun 2022 – Jul 2023*
Master's Thesis: Reinforcement Learning for Battery Charging Algorithms; Guide – Prof. R Krishna Kumar
 - Developed a model of an LFP cell in **MATLAB Simulink** to simulate the electrical, thermal, and aging characteristics after conducting an extensive literature survey on the modelling of Li-ion cells
 - Designed a Proximal Policy Optimisation based RL algorithm in **MATLAB** for the fast charging of the cell, while ensuring **fast charging in an ambient temperature range of 10 °C - 40 °C**
 - Achieved a **50% reduction** in charging time at the same ambient temperature when compared to 1C CCCV charging, **while maintaining a similar battery degradation**
 - The abstract was selected for the **Cyber Physical Systems Summit 2023**, held at IIT Kharagpur ( )
- **Researcher** – Young Research Fellow Program, IITM |  *Sep 2020 – Jun 2021*
Obstacle Avoidance and Target Driven Guidance Algorithm; Guide – Prof. Satadal Ghosh
 - Designed a Hybrid **Obstacle Avoidance and Target Driven Guidance algorithm** in a 2D field for unmanned vehicles, incorporating proportional navigation and acceleration-velocity obstacle algorithms

- Simulated the UAV in a MATLAB environment with constant velocity obstacles and observed a **5% reduction in the time taken to reach the target** over other classical aerospace approaches

PROJECTS

- **Team Anveshak** – Centre for Innovation, IITM, Chennai, India |  *Feb 2019 – May 2020*
 - Developed a **rocker-bogie suspension** using carbon fibre linkages, customising the link lengths using MATLAB to minimize chassis vibration when travelling over a sine wave
 - Designed and developed a new wheel for an autonomous rover, and achieved a **5% weight reduction** by using custom 3D printed TPU tyres, while making the wheel **10% larger**
 - Designed a **modular, low-cost onboard science laboratory** for collection and analysis of soil samples for presence of biomarkers
 - Achieved **5th position** in Indian Rover Challenge 2020, and ranked 8th in the design review phase of International Rover Challenge 2020, qualifying for the finals
- **Anti-Lock Brake System Modelling** – Course Project: Vehicle Dynamics *Jan 2021 – May 2021*
 - Developed a State-Space model based on vehicle and tyre parameters in MATLAB and a bicycle model in Simulink
 - Designed PID and Bang-Bang controller for **Anti-lock Braking System (ABS)** for the bicycle model
 - Evaluated yaw rate response and lateral acceleration response with Mimuro Plot with varying tyre parameters
- **Sliding Mode Control for Tractor Hitch Control** – Course Project: Controls of Automotive Systems *Nov 2021*
 - Modelled the electro-hydraulic system of a tractor in a controller-canonical form derived from a given transfer function
 - Reduced chattering by implementing a sliding mode control with a super-twisting algorithm
- **IoT Based Interactive Game** – Course Project: Smart Sensing for IoT *July 2021 – Nov 2021*
 - Developed a virtual tennis game for Windows using Unity, which used an Android phone as a tennis racket by utilising the IMU sensor data
 - Developed a random forest model to classify gestures into four different types of shots as input for the game
- **Case Study: Use of Mobile Robotics in Industries** – Course Project: Industry 4.0 and Smart Sensing *May 2023*
 - Conducted a case study on the application of multiple types of automated mobile robotics across varying industries, analysing their implementation, cost, and impact

TECHNICAL SKILLS

Software: SolidWorks, Simulink, ImageJ, Blender

Languages: Python, MATLAB, C, C++, Mathematica

POSITIONS OF RESPONSIBILITY AND VOLUNTEER WORK

- **Teaching Assistant** – Electromagnetic Compatibility (EMC) for Product Design *Jul 2022 – Nov 2022*
 - Instructed undergraduate students in operating oscilloscopes and analysing electronic signals as part of the EMC laboratory; evaluated lab assignments and provided feedback to students
- **Department Legislator & Placement Coordinator** – Dept. of Engineering Design, IITM *Jul 2020 – May 2021*
 - Acted as a representative for the student body of the Dept. of Engineering Design
 - Resolved student-wide concerns as a legislator in the Student Legislative Council, including establishing a COVID-19 grievance committee
 - Coordinated with **5+ companies** to organize sessions on the relevance of Automotive Engineering, **resulting in internship and full-time hiring opportunities for students**

MISCELLANEOUS

- **Scholastic Achievements**
 - Received the K.C. Mahindra Scholarship for Post Graduate studies in 2026
 - Secured a rank of 2318 (top 0.2%) in JEE Main 2018, and a rank of 3486 (top 1.5%) in JEE Advanced 2018
 - Received the Kishore Vaigyanik Protsahan Yojana (KVPY) Scholarship in 2017, with a rank of 1226
- **Interests**
 - Lego, Swimming, Running, Motorcycling, Gaming, Football, Cricket